

Graphs of Sine and Cosine Functions

ANSWER KEY



Amplitude, period, and midline

- 1 For the function $y = 3\sin(2x)$, state:
a the amplitude $= 3$ **b** the period $= \pi$ **c** the range $[-3, 3]$
 - 2 For the function $y = -2\cos\left(\frac{x}{2}\right) + 1$, state:
a the amplitude $= 2$ **b** the period $= 4\pi$
c the midline $y = 1$ **d** the range $[-1, 3]$
 - 3 For $y = \sin\left(x - \frac{\pi}{3}\right)$, state the phase shift and the period.
Phase shift: $\frac{\pi}{3}$ to the right. Period: 2π .
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Reading and writing equations from graphs

- 4 The graph below shows one sinusoidal function. State its amplitude and period, then write an equation for the function.
Amplitude 2, period 2π . Equation: $y = 2\sin x$.
 - 5 Write an equation of a cosine function with amplitude 4, period π , reflected across the x -axis, with no phase or vertical shift.
 $y = -4\cos(2x)$ (period $\frac{2\pi}{b} = \pi$ gives $b = 2$).
 - 6 The depth of water at a harbour entrance is modeled by $d(t) = 2.5\sin\left(\frac{\pi}{6}t\right) + 7$ metres, where t is in hours after midnight.
a State the maximum and minimum depths. Maximum $7 + 2.5 = 9.5$ m; minimum $7 - 2.5 = 4.5$ m.
b State the period of the tide cycle. Period $= \frac{2\pi}{\pi/6} = 12$ hours.
c Find the depth at $t = 3$. $d(3) = 2.5\sin\left(\frac{\pi}{2}\right) + 7 = 9.5$ m.
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Graphing with transformations

- 7 The graph below shows $y = \cos(x) + 1$, a cosine curve with a vertical shift. Identify the midline and the maximum and minimum values shown.
Midline $y = 1$; maximum value 2; minimum value 0.
- 8 Describe, in order, the transformations that turn $y = \sin x$ into $y = -3\sin\left(2\left(x - \frac{\pi}{4}\right)\right) + 2$.
Horizontal compression by a factor of 2 (period π), reflection across the x -axis, vertical stretch by a factor of 3, shift right $\frac{\pi}{4}$, and shift up 2.

- 9 State the amplitude, period, phase shift, and midline for $y = 5\cos\left(3x + \frac{\pi}{2}\right) - 4$. (Hint: factor the argument as $3\left(x + \frac{\pi}{6}\right)$.)
Amplitude 5; period $\frac{2\pi}{3}$; phase shift $\frac{\pi}{6}$ to the left; midline $y = -4$.
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Modeling periodic phenomena

- 10 The average monthly temperature in a city varies between 8°C (in January, month 1) and 28°C (in July, month 7), following a sinusoidal pattern with a 12-month period.
- a Find the amplitude and midline of the model. Amplitude $= \frac{28-8}{2} = 10$; midline $= \frac{28+8}{2} = 18$.
- b Write a cosine model $T(m)$ for the temperature in month m , using the fact that the minimum occurs at $m = 1$. Since the minimum is at $m = 1$, use a negative cosine shifted:
$$T(m) = -10\cos\left(\frac{2\pi}{12}(m-1)\right) + 18 = -10\cos\left(\frac{\pi}{6}(m-1)\right) + 18.$$
- c Use your model to estimate the temperature in month 4 (April).
$$T(4) = -10\cos\left(\frac{\pi}{6}(3)\right) + 18 = -10\cos\left(\frac{\pi}{2}\right) + 18 = 18^\circ\text{C}.$$